

# Questions for the Audience

## Which Elements are Required to Define a Loss Function?

- A dataset with inputs  $X$  and targets  $T$
- A model to produce an output for an input  $y = f(x)$
- A comparison function between output and target  $\text{Cmp}(y, t)$

## What Kinds of Problems can Arise in Gradient Descent

- A local minimum is reached
- We get stuck on a plateau of the loss surface
- We oscillate or diverge, i.e., the loss increases
- We end up in a saddle point

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Assume a Linear Model  $f_{\vec{w}}$  with  $D + 1$  Parameters

- How many inputs can this model have?
  - $D$  inputs, when the first weight  $w_0$  belongs to the bias
  - $D + 1$  inputs if we have no bias, but then  $f_{\vec{w}}(\vec{0}) = 0$
- How many values does the gradient  $\nabla_{\vec{w}}$  have?
  - $D + 1$  elements, one for each weight
- Which kinds of applications can we perform with this model?
  - Regression
  - Binary classification (requires activation function)